

Smart Renewable Energy Microgrid

A small team of enthusiastic and adventurous engineers at FluxGen Engineering Technologies has come up with a solution to electrify houses in remote rural areas that are not directly connected to the electricity board power grid. Read on to know more

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Microgrid is basically a small-scale power supply network that is designed to provide power for a small community. It cannot be used for high-power consuming devices but can be used as an alternative approach to integrate small-scale distributed energy resources into low-voltage electricity systems. Enabling local power generation, it comprises various small power generating sources that make it highly flexible and efficient.

Basically, the solution aims to electrify houses that are not directly connected to the electricity board power grid due to their remoteness. To be precise, "We have integrated existing hardware components in the market with a powerful embedded system. The setup forms an electricity grid that is small enough to be called a microgrid. This system is powered by renewable energy sources such as solar photovoltaics (PV), wind and micro-hydel. It can be remotely monitored from anywhere in the world," explains Ganesh Shankar, managing director, FluxGen Engineering Technologies.

Fig. 1 shows the model of the smart renewable energy microgrid.

From concept to product

"Over 400 million Indians either do not have access to electricity or have intermittent access to electricity. Connecting the whole country to the main electricity grid is a task that would take several years and in some places it may not make an economic case," points out Hari D.K., chief engineer, FluxGen Engineering Technologies. Distributed energy generation and distribution can



Fig. 1: Models of houses which form part of the smart microgrid setup at FluxGen Lab



Fig. 2: National Instruments' Single Board RIO (sbRIO 9641) used to intelligently control and monitor the smart microgrid

potentially solve this problem.

An off-the-grid renewable energy plant connected to a network of houses in rural areas will potentially provide them the same quality of power as obtained from the main electricity grid.

"In India, electricity grid penetration is very poor. Fortunately, the communication network penetration in

India is much higher. This makes the case for a smart microgrid whose performance can be remotely monitored and virtually supervised," says Shankar.

"We are planning to put our systems in places where there is no electricity. But in case the grid does penetrate, the microgrid should ideally be compatible with the grid for the following reasons:

1. It could optimally utilise grid power and renewable energy power, ensuring that the customers get electricity most of the time.
2. If power generated from the renewable energy source is in excess, it should be able to feed it back to the grid so that other microgrids connected to the network could utilise it," he said.

Regarding the time taken in developing their concept to product, Shankar says, "Before starting FluxGen I was working with GE. There I happened to work on building smart energy meters for about six months. During that time, I got a good under-